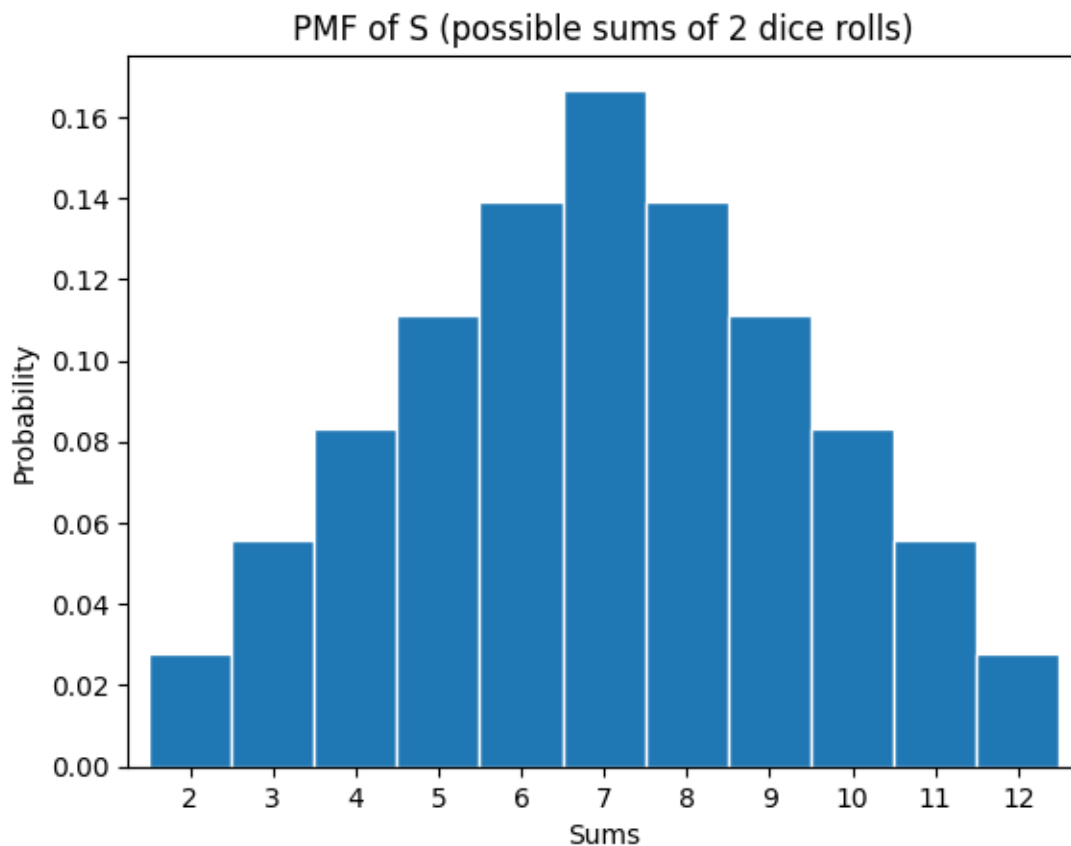


0.1 Question 2e (1 point)

Plot a histogram of the PMF of S . (Use the same syntax showed above so that the bin widths are equal to 1, there's a white lines between each value and we can interpret area in the histogram as probabilities. Be sure to label the title and axes.)

```
In [67]: s_pmf.plot.bar(rot=0, width=1, ec='white')
plt.title("PMF of S (possible sums of 2 dice rolls)")
plt.xlabel("Sums")
plt.ylabel("Probability")
```

```
Out[67]: Text(0, 0.5, 'Probability')
```



0.2 Question 2f (2 points)

Are events $D_1 = 1$ and $D_2 = 1$ independent or dependent? Justify your answer using the mathematical definition of independence.

Write up your full solution in the box below using LaTeX (not code). Show all steps fully justifying your answers.

If D_1 and D_2 are independent, then $P(D_1 \cup D_2) = P(D_1)P(D_2)$ holds.

$$P(D_1 \cup D_2) = P(D_1)P(D_2) \tag{1}$$

$$\frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} \tag{2}$$

$$\frac{1}{36} = \frac{1}{36} \tag{3}$$

Therefore D_1 and D_2 are independent.

0.3 Question 2g (2 points)

Are events $D_1 = 3$ and $S = 7$ independent or dependent? Justify your answer using the mathematical definition of independence.

Write up your full solution in the box below using LaTeX (not code). Show all steps fully justifying your answers.

Using the same definition as before:

$$P(D_1 \cup S) = P(D_1)P(S) \quad (4)$$

The only way S and D_1 can both be true is if D_2 is 4. Thus $P(D_1 \cup S) = \frac{1}{36}$. Additionally, $P(S) = \frac{1}{6}$ from the PMF.

$$\frac{1}{36} = \frac{1}{6} \cdot \frac{1}{6} \quad (5)$$

$$\frac{1}{36} = \frac{1}{36} \quad (6)$$

Therefore D_1 and S are independent.

0.4 Question 2h (2 points)

Are events $D_1 = 1$ and $S = 6$ independent or dependent? Justify your answer using the mathematical definition of independence.

Write up your full solution in the box below using LaTeX (not code). Show all steps fully justifying your answers.

Using the same definition as before:

$$P(D_1 \cup S) = P(D_1)P(S) \tag{7}$$

The only way S and D_1 can both be true is if D_2 is 5. Thus $P(D_1 \cup S) = \frac{1}{36}$. Additionally, $P(S) = \frac{5}{36}$ from the PMF.

$$\frac{1}{36} = \frac{1}{6} \cdot \frac{5}{36} \tag{8}$$

$$\frac{1}{36} \neq \frac{11}{36} \tag{9}$$

Therefore D_1 and S are dependent.

0.5 Question 2i (2 points)

Are the **random variables** D_1 and S independent or dependent? Justify your answer using the definition of independent random variables.

Write up your full solution in the box below using LaTeX (not code). Show all steps fully justifying your answers.

Let $d \in D_1$ and $s \in S$. D_1 and S are independent if the following is true:

$$P(D_1 = d, S = s) = P(D_1 = d)P(S = s) \quad \forall d, s \quad (10)$$

From the previous question, if $d = 1$ and $s = 6$, then D_1 and S are dependent. Since all d and s don't result in independent events, the random variables are dependent.

0.6 Question 3a (3 points)

We define **net winnings** to be the amount you win minus the amount you paid to play.

Let X be a random variable that gives your **net winnings if you bet on red** and the roulette wheel is **spun once**.

i). What is the probability distribution (aka Probability Mass Function) of X ? Give your answer as a Markdown table.

- Tip: To create Markdown tables that will compile when you create the PDF in this assignment, please refer to this syntax guide: <https://www.markdownguide.org/extended-syntax/#tables>. Make sure you have a blank line before starting any tables, otherwise they won't compile correctly in the PDF.

ii). Calculate the **expected value** of your net winnings by betting on red.

iii). Calculate the **variance** of your net winnings by betting on red.

Write up your full solution to all 3 parts in the SAME box below using Markdown and LaTeX (not code). Show all steps fully justifying your answer.

i)

Let R be

Value	Probability
1	$\frac{18}{38}$
-1	$\frac{20}{38}$

ii)

$$\mathbb{E}[R] = \left(1 \cdot \frac{18}{38}\right) + \left(-1 \cdot \frac{20}{38}\right) \quad (11)$$

$$= \frac{18}{38} - \frac{20}{38} \quad (12)$$

$$= -\frac{2}{38} = -\frac{1}{19} \quad (13)$$

iii)

$$Var(R) = \mathbb{E} [R^2] - \mathbb{E} [R]^2 \quad (14)$$

$$= \left[1^2 \cdot \frac{18}{38} + (-1)^2 \cdot \frac{20}{38} \right] - \left(-\frac{1}{19} \right)^2 \quad (15)$$

$$= \left(\frac{18}{38} + \frac{20}{38} \right) - \frac{1}{361} \quad (16)$$

$$= \frac{360}{361} \quad (17)$$

0.7 Question 3c (3 points)

In Roulette you can bet on one of three “dozens” segments, called 1st 12, 2nd 12, and 3rd 12. They cover 1-12, 13-24, and 25-36, respectively. If you bet \$1 on the first dozen (or second dozen, or third dozen) nonzero numbers and win, then your **net winnings** is \$2 (i.e. you get your original dollar back, plus another \$2).

Let Y be a random variable that gives your **net winnings** if you bet on any one of the three “dozen” nonzero numbers and the roulette wheel is spun once.

i). What is the probability distribution of Y ? Give your answer as a table.

ii). What is $E[Y]$?

iii). What is $Var[Y]$?

Write up your full solution to all 3 parts above in the SAME box below using LaTeX (not code). Show all steps fully justifying your answer.

i)

Value	Probability
2	$\frac{12}{38}$
-1	$\frac{26}{38}$

ii)

$$\mathbb{E}[Y] = \left(2 \cdot \frac{12}{38}\right) + \left(-1 \cdot \frac{26}{38}\right) \quad (18)$$

$$= -\frac{1}{19} \quad (19)$$

iii)

$$Var(Y) = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 \quad (20)$$

$$= \left[2^2 \cdot \frac{12}{38} + (-1)^2 \cdot \frac{26}{38}\right] - \left(-\frac{1}{19}\right)^2 \quad (21)$$

$$= \frac{702}{361} \quad (22)$$

0.8 Question 3e (3 points)

Recall, we showed in class that the expected value of your net winnings if you bet on any number is also $\$ - \frac{1}{19}$.

So you're hopefully onto the pattern by now. The net winnings in Roulette are designed so that the expected value of the net winnings is always $-\frac{1}{19}$ (i.e. on average the gambler loses money and the casino makes money).

Since we define these in terms of your net winnings after betting \$1, we can think of these as net winning odds.

For example, since if you bet \$1 on the first dozen nonzero numbers and win, then your net winnings are \$2, we say the odds are 2 to 1 (denoted 2:1).

The odds are 35:1 for landing on any particular number. This means if you bet \$1, your net winnings will be \$35.

Suppose the casino wanted to develop odds for a new bet in Roulette, where they allow you to bet on any set of 3 different numbers. Let the odds for this new bet be

$$x : 1$$

(This means if you bet \$1, your net winnings will be \$ x).

What should \$ x be so that the expected value of the net winnings is still $-\frac{1}{19}$?

Show your work using LaTeX below.

Value	Probability
x	$\frac{3}{38}$
-1	$\frac{35}{38}$

Let W be winning the bet.

$$\mathbb{E}[W] = -\frac{1}{19} = \left(x \cdot \frac{3}{38}\right) + \left(-1 \cdot \frac{35}{38}\right) \quad (23)$$

$$-\frac{1}{19} = x \cdot \frac{3}{38} - \frac{35}{38} \quad (24)$$

$$\frac{33}{38} = x \cdot \frac{3}{38} \quad (25)$$

$$x = 11 \quad (26)$$

0.9 Question 3f (2 points)

Let's generalize this!

Define a function $x(n)$ that describes the odds the casino should give for betting \$1 on any set of n numbers if the casino wants to keep the expected value of the net winnings at $-\frac{1}{19}$ for $0 < n \leq 36$. (For example, the odds for betting on any 3 different numbers should be set at $x(3)$ to 1. The odds for betting on any 4 different numbers should be set at $x(4)$ to 1).

Write your solution in the cell below:

From the previous question:

Value	Probability
$x(n)$	$\frac{n}{38}$
-1	$\frac{38-n}{38}$

$$\mathbb{E}[W] = -\frac{1}{19} = \left(x(n) \cdot \frac{n}{38}\right) + \left(-1 \cdot \frac{38-n}{38}\right) \quad (27)$$

$$-\frac{1}{19} = x(n) \cdot \frac{n}{38} - \frac{38-n}{38} \quad (28)$$

$$x(n) \cdot \frac{n}{38} = -\frac{1}{19} + \frac{38-n}{38} \quad (29)$$

$$x(n) = -\frac{2}{n} + \frac{38-n}{n} \quad (30)$$

$$x(n) = \frac{36-n}{n} \quad (31)$$

For example, $x(3) = 11$

0.10 Question 4a (8 points)

Answer all of the parts below in the SAME cell below using LaTeX. Show all of your steps.

i). Determine the value of a such that this defines a valid probability distribution. Use that value for the rest of the problem.

ii). Calculate $P(X \leq 3)$.

iii). What is $E[X]$? (Show steps calculating this).

iv). What is the standard deviation of X ? (Show all steps calculating this).

Answer all of the parts above in SINGLE cell provided below using LaTeX.

i)

In order for the distribution to be valid, all values must add to 1, so

$$2a(2)^2 - 2a(2) + 2a(3)^2 - 2a(3) + 2a(4)^2 - 2a(4) = 1 \quad (32)$$

$$2a(4) - 2a(2) + 2a(9) - 2a(3) + 2a(16) - 2a(4) = 1 \quad (33)$$

$$8a - 4a + 18a - 6a + 32a - 8a = 1 \quad (34)$$

$$40a = 1 \quad (35)$$

$$a = \frac{1}{40} \quad (36)$$

The probabilities for each k is thus:

$$P(X = 2) = \frac{8 - 4}{40} = \frac{1}{10} \quad (37)$$

$$P(X = 3) = \frac{18 - 6}{40} = \frac{3}{10} \quad (38)$$

$$P(X = 4) = \frac{32 - 8}{40} = \frac{6}{10} \quad (39)$$

ii)

$$P(X = 3) + P(X = 4) = \frac{4}{10} = \frac{2}{5} \quad (40)$$

iii)

$$\mathbb{E}[X] = 2 \cdot P(X = 2) + 3 \cdot P(X = 3) + 4 \cdot P(X = 4) \quad (41)$$

$$= 2 \cdot \frac{1}{10} + 3 \cdot \frac{3}{10} + 4 \cdot \frac{6}{10} \quad (42)$$

$$= \frac{2}{10} + \frac{9}{10} + \frac{24}{10} \quad (43)$$

$$= \frac{35}{10} = 3.5 \quad (44)$$

iv)

$$Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = [2^2 \cdot P(X = 2) + 3^2 \cdot P(X = 3) + 4^2 \cdot P(X = 4)] - 3.5^2 \quad (45)$$

$$= \left[2^2 \cdot \frac{1}{10} + 3^2 \cdot \frac{3}{10} + 4^2 \cdot \frac{6}{10} \right] - 3.5^2 \quad (46)$$

$$= 0.45 \quad (47)$$

Since the standard deviation is $\sqrt{Var(X)}$, it is about 0.67082.

0.11 Question 4b (2 points)

Plot a histogram of the discrete probability distribution for X .

Use the same plotting guidelines as shown in Problem 2: - Set the bin widths to be equal to 1 - Add white lines between each bar

Be sure to include a title and label your x and y axes on your plot.

```
In [77]: dist = pd.Series([1/10, 3/10, 6/10], index=[2, 3, 4])
```

```
dist.plot.bar(rot=0, width=1, ec='white')
plt.title("Probability distribution for X")
plt.xlabel("k")
plt.ylabel("Probability")
# Your code for the histogram above this line
```

```
Out[77]: Text(0, 0.5, 'Probability')
```

